`

HYBRID SENSOR-BASED EARLY DETECTION OF ROADSIDE HUMANS AND ANIMALS FOR ACCIDENT PREVENTION

A project submitted to the Bharathidasan University
in partial fulfillment of the requirements
for the award of the Degree of

BACHELOR OF COMPUTER APPLICATIONS

Submitted by

R.IRFAN

Register Number: 235113514

Under the guidance of

Dr. K. ADLIN SUJI, M.C.A, M.Phil, Ph.D., NET.,

Assistant Professor

DEPARTMENT OF COMPUTER APPLICATIONS

BISHOP HEBER COLLEGE (AUTONOMOUS)

"Nationally Reaccredited at A^{++} Grade with a CGPA of 3.69 out of 4 in NAAC IV Cycle"

Recognized by UGC as "College of Excellence"
Affiliated to Bharathidasan University

TIRUCHIRAPPALLI-620 017

OCTOBER – 2025

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DECLARATION

I hereby declare that the project work presented is originally done by me under the guidance of **Dr.K.ADLIN SUJI, M.C.A., M.Phil., Ph.D., NET**., Assistant Professor, UG Department of Computer Applications, Bishop Heber College (Autonomous), Tiruchirapalli-620017 and has not been include in any thesis/project submitted for any other degree

Name of the Candidate. : IRFAN R

Register Number : 235113514

Batch : 2023-2026

Signature of the Candidate

`

Dr. K. ADLIN SUJI, M.C.A., M.Phil., Ph.D., NET.,
Assistant Professor,
UG Department of Computer Applications,
Bishop Heber College (Autonomous),
Tiruchirappalli – 620017

Date:

CERTIFICATE

This is to certify that the project work entitled **"Hybrid Sensor-Based Early Detection of Roadside Humans and Animals For Accident Prevention"** is a bonafide record work done by **R.Irfan, Register Number: 235113514** in partial fulfilment of the requirements for the award of the degree of **BACHELOR OF COMPUTER APPLICATIONS** during the period **2023-2026**

Place: Trichy **Signature of the Guide**

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UG DEPARTMENT OF COMPUTER APPLICATIONS

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TIRUCHIRAPPALLI – 620 017

Date:

Course Title: Project **Course Code: U23CA5PJ**

CERTIFICATE

The Viva-Voce examination for the candidate **R.IRFAN, Register Number: 235113514** was held on _____.

Signature of the Guide **Signature of the HOD**

Examiners:

1.

2.

`

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ABSTRACT

Road accidents due to the sudden presence of animals or humans on highways and urban roads have become a major safety concern worldwide. Many drivers fail to react in time when unexpected obstacles appear, leading to severe collisions, injuries, and fatalities. Existing safety systems are limited in providing real-time detection and instant alerts to prevent such accidents. To address this challenge, the proposed project integrates Artificial Intelligence (AI) with the Internet of Things (IoT) to build a smart road safety mechanism. The system utilizes AI-based image processing to detect animals and humans on the road with high accuracy. When a potential obstacle is identified near the vehicle, a voice alert is immediately triggered to warn the driver, allowing timely action to avoid collisions. In case an accident occurs despite the warning, the IoT module sends an automated SMS containing the vehicle's location to the registered family members, ensuring rapid emergency communication. This dual functionality of preventive alerting and post-accident reporting enhances road safety, reduces accident risks, and provides timely assistance during emergencies. By combining AI-driven detection with IoT-based communication, the system offers a reliable and efficient solution to minimize road hazards caused by unexpected human or animal presence.

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INTRODUCTION

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1. INTRODUCTION

Road safety remains a critical concern worldwide due to the increasing number of vehicles, urban expansion, and diverse environmental challenges. One of the major causes of road accidents is the unexpected appearance of pedestrians and animals on roadways, which often leads to collisions and fatalities. With the rapid growth of traffic density and the complexity of urban and rural roads, traditional measures like signage, fences, and traffic signals are often insufficient to prevent accidents. As a result, there is an urgent need for automated systems that can monitor road environments in real-time and alert drivers to potential hazards, thereby reducing accident rates and saving lives.

Fig 1.1 Animal detection

Animals crossing roads, particularly in rural and suburban areas, pose a significant threat to vehicle occupants and the animals themselves. In many countries, collisions with wildlife, livestock, and stray animals contribute to substantial economic losses, property damage, and human injuries. Similarly, pedestrian-related accidents are a global public health issue, often resulting in severe injuries or fatalities. Children, elderly individuals, and people with disabilities are especially vulnerable when crossing streets without proper safeguards. In this context, developing technologies that can effectively detect the presence of humans and animals in or near roadways has become a crucial area of research in intelligent transportation systems.

Modern advancements in sensors, computing, and communication technologies have opened new possibilities for real-time road monitoring. Computer vision, artificial intelligence, and deep learning algorithms allow for the automatic detection and classification of objects, including pedestrians, cyclists, and animals. Vision-based systems using cameras

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can capture road scenes and analyze them to identify potential hazards, even under challenging conditions like low light or bad weather. Additionally, sensor fusion approaches, which combine inputs from cameras, LiDAR, ultrasonic sensors, and radar, enhance detection accuracy and reliability. By processing these signals in real-time, intelligent systems can alert drivers or even autonomously control vehicles to avoid collisions.

The integration of Internet of Things (IoT) technology into road safety systems further enhances monitoring capabilities. IoT-enabled devices can transmit information about detected animals or pedestrians to nearby vehicles, traffic control centers, or emergency services. For instance, when a deer or stray dog is detected near a highway, the system can send instant alerts to approaching vehicles, reducing the risk of accidents. Similarly, in urban areas, the presence of pedestrians near busy intersections can trigger alerts or activate automated braking systems in smart vehicles. These technologies not only improve reaction times but also contribute to the broader goals of smart city infrastructure and autonomous transportation systems.

Despite these advancements, several challenges remain in effectively detecting animals and humans on roads. Variability in object appearance, unpredictable movement patterns, and environmental factors such as rain, fog, or night-time darkness can hinder detection accuracy. Moreover, different species of animals vary widely in size, color, and behavior, making it difficult for a single detection system to account for all scenarios. Pedestrian detection also faces challenges, especially in crowded urban environments where occlusions and high-density traffic complicate reliable identification. To address these issues, researchers are continually developing more robust machine learning models, training datasets with diverse scenarios, and algorithms capable of adaptive learning under dynamic road conditions.

The importance of animal and pedestrian detection extends beyond immediate safety concerns. Reducing road accidents also has significant social, economic, and environmental implications. Fewer accidents mean lower medical costs, reduced vehicle repair expenses, and decreased traffic congestion. Preventing collisions with wildlife preserves biodiversity and reduces the ecological impact of transportation networks. Furthermore, smart detection systems contribute to the advancement of autonomous vehicles, which rely heavily on accurate perception of the surrounding environment to make safe navigation decisions. As autonomous driving technology becomes more widespread, integrating reliable animal and

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pedestrian detection mechanisms will be critical to achieving fully autonomous and safe transportation systems.

In detecting animals and humans on roads is a vital aspect of modern intelligent transportation systems aimed at enhancing road safety and reducing accidents. Advances in computer vision, AI, and IoT technologies provide promising solutions for real-time detection and proactive alerting of potential hazards. While challenges remain in terms of environmental variability, unpredictable movements, and diverse object appearances, ongoing research continues to improve the accuracy and reliability of these systems. Implementing such technologies has far-reaching implications for public safety, economic savings, environmental protection, and the future of autonomous mobility. As road networks continue to expand and traffic density increases, the development and deployment of effective animal and pedestrian detection systems

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EXISTING SYSTEM AND PROPOSED SYSTEM

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2. EXISTING SYSTEM AND PROPOSED SYSTEM

2.1 EXISTING SYSTEM

The animal and human detection on roads mainly rely on camera-based monitoring and vehicle-integrated assistance technologies that aim to minimize accidents. Roadside surveillance cameras and vehicle-mounted vision systems are used to capture real-time road activity, enabling the identification of pedestrians or larger animals crossing unexpectedly. Advanced driver assistance systems (ADAS), such as Automatic Emergency Braking (AEB) and Pedestrian Detection, utilize image processing and machine learning techniques to classify humans and animals directly in the vehicle's path, allowing vehicles to respond automatically by slowing down or stopping. While these approaches have improved safety in urban and highway environments, they remain limited in scope as they often struggle with small animals, unpredictable movements, or poor visibility conditions. Moreover, many existing systems function only within specific vehicle models or controlled environments, and they lack wider connectivity to ensure broader, collective safety. Consequently, current solutions are helpful but still fall short of offering fully reliable and adaptable protection for both pedestrians and animals across varying traffic and environmental conditions.

2.1.1 DISADVANTAGES

- Limited accuracy in low-visibility conditions.
- Struggles to detect small or fast animals.
- Works only in specific vehicle models.
- Lacks wide connectivity for real-time safety.

2.2 PROPOSED SYSTEM

The proposed system introduces an AI and IoT-based road safety mechanism designed to detect the sudden presence of humans or animals near a moving vehicle and provide timely alerts to the driver. The system uses AI-powered object detection through cameras or sensors to continuously monitor the road environment and identify obstacles in real time. Once an

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animal or human is detected, a voice alert is instantly generated to warn the driver, allowing them to take preventive action and avoid collisions. In addition, the IoT module ensures accident management by automatically sending an SMS notification containing the vehicle's GPS location to the registered family members if a crash occurs. This integration of AI-based detection with IoT-enabled communication not only improves driver awareness but also provides immediate support during emergencies, making the system a comprehensive solution for enhancing road safety and minimizing accident-related risks.

2.2.1 ADVANTAGES

- Real-time detection of animals and humans.
- Instant voice alert to warn drivers.
- Automated accident notification via SMS.
- GPS location sharing for quick response.
- Enhanced road safety and risk reduction.

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HARDWARE REQUIREMENTS AND

SOFTWARE REQUIREMENTS

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3. HARDWARE REQUIREMENTS AND SOFTWARE REQUIREMENTS

3.1 Hardware Requirements

- ESP 32 microcontroller
- Vibration sensor
- Web cam
- PIR sensor
- Ultrasonic sensor
- GSM
- GPS
- LCD
- Power supply
- Buzzer
- LEDs

a) Software Requirements

- Ardino IDE
- Python

b) Hardware Description

3.1.1 ESP32 Microcontroller

The ESP32 microcontroller, developed by Espressif Systems, is a highly versatile and powerful microcontroller platform that has gained significant popularity in the fields of embedded systems, Internet of Things (IoT), and real-time applications. Designed as a successor to the well-known ESP8266, the ESP32 offers improved performance, higher efficiency, and a broad set of features that make it suitable for a wide range of projects, from

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simple sensor data collection to advanced robotics and automation systems. Built with a dual-core or single-core Tensilica Xtensa processor, the ESP32 operates at speeds up to 240 MHz, providing exceptional computational power in a compact and low-cost package. In addition, it integrates both Wi-Fi and Bluetooth connectivity, enabling seamless wireless communication and making it one of the most adaptable microcontrollers available today.

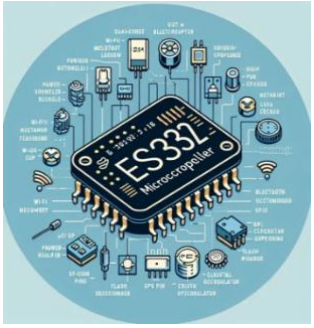

Fig 3.1 ESP 32 Microcontroller

One of the most significant advantages of the ESP32 is its connectivity capabilities. Unlike many traditional microcontrollers that require external modules for wireless communication, the ESP32 comes with built-in Wi-Fi (802.11 b/g/n) and Bluetooth (classic and BLE). This dual connectivity allows it to function in various roles, including as a web server, IoT hub, or device controller. For IoT applications, the ESP32 can directly connect to cloud platforms and services, enabling real-time data transmission and remote monitoring without the need for additional networking hardware. Bluetooth Low Energy (BLE) also makes it particularly useful in wearable devices and applications where low-power consumption is critical. These features have positioned the ESP32 as an essential component in modern IoT ecosystems.

In terms of hardware features, the ESP32 is designed with flexibility in mind. It provides a wide variety of peripherals such as multiple General-Purpose Input/Output (GPIO) pins, Analog-to-Digital Converters (ADC), Digital-to-Analog Converters (DAC), capacitive touch sensors, and support for communication protocols like I^2C, SPI, UART, CAN, and PWM. This rich peripheral set allows developers to interface with sensors, actuators, and external devices easily, thereby simplifying the design of complex embedded applications. Additionally, its ultra-low-power co-processor enables the microcontroller to perform basic

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tasks while in deep sleep mode, drastically reducing energy consumption and extending battery life for portable and energy-sensitive devices. These capabilities make the ESP32 especially valuable for applications where energy efficiency and long operational lifetimes are required.

Another noteworthy aspect of the ESP32 is its development ecosystem and support. Espressif provides an official development framework called the ESP-IDF (IoT Development Framework), which allows developers to write applications in C and C++. For those who prefer higher-level programming, the ESP32 is also compatible with the Arduino IDE, MicroPython, and PlatformIO, making it accessible to both beginners and advanced developers. The availability of comprehensive documentation, an active online community, and open-source libraries further enhance its usability. This wide development support ensures that users can easily find resources, tutorials, and solutions for troubleshooting, which accelerates the process of prototyping and project deployment.

From an application perspective, the ESP32 is incredibly versatile. It is used in smart home automation systems to control lighting, security, and appliances through wireless commands. In industrial automation, it plays a role in monitoring machines, collecting sensor data, and communicating with centralized systems for predictive maintenance and efficiency improvements. In healthcare, ESP32 modules are integrated into wearable devices to track vital parameters and transmit data to medical professionals. Its role extends to robotics, drones, environmental monitoring, and even educational purposes where students learn programming and embedded systems through hands-on experimentation. The combination of processing power, wireless communication, and low cost makes it an attractive choice for diverse domains.

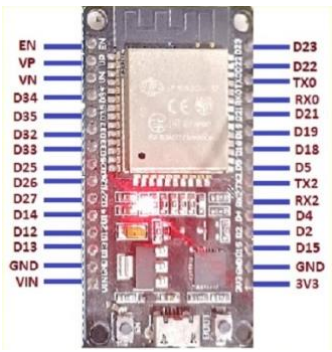

Fig 3.2 Pin details

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The ESP32 also emphasizes security features, which are critical in IoT applications. It supports hardware encryption, secure boot, and flash encryption, ensuring that sensitive data remains protected during communication and storage. With cybersecurity becoming a major concern in modern technology, these built-in safeguards allow developers to create secure IoT applications without the need for complex external measures. Furthermore, the chip is designed to handle over-the-air (OTA) updates, allowing devices to be updated remotely with new firmware, thereby ensuring long-term usability and adaptability to changing needs.

In terms of performance, the ESP32 delivers remarkable efficiency for its size. With its dual-core processor architecture, it can perform multitasking effectively, allowing different processes such as data collection, processing, and communication to run simultaneously. Its clock speed and integrated peripherals ensure that it can handle computationally intensive tasks such as real-time signal processing and machine learning inference on-device. The inclusion of large memory capacity, both in terms of SRAM and external flash support, further enhances its ability to manage complex applications.

In the ESP32 microcontroller represents a breakthrough in the microcontroller domain by integrating high performance, wireless connectivity, energy efficiency, and robust development support into a single platform. Its wide range of features makes it suitable for applications ranging from hobbyist DIY projects to advanced industrial and medical systems. By enabling developers to create powerful, secure, and scalable solutions at a relatively low cost, the ESP32 has established itself as one of the most influential microcontrollers in the era of IoT and smart technology. Its continuing popularity and active ecosystem ensure that it will remain a cornerstone of embedded system development for years to come.

3.1.2 Vibration Sensor

A vibration sensor is an essential component in modern sensing and monitoring systems that detects oscillations, disturbances, or movements of an object or environment. It works on the principle of converting mechanical vibrations into corresponding electrical signals, which can then be processed for analysis or decision-making. These sensors come in various forms such as piezoelectric sensors, accelerometers, and MEMS-based devices, each optimized for different levels of sensitivity and application areas. The core function of a vibration sensor is to measure the intensity, frequency, and pattern of vibrations. This capability is vital because vibration often serves as an early indicator of mechanical faults,

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structural weaknesses, or external impacts. By analyzing vibration data, engineers and researchers can prevent failures, enhance efficiency, and maintain safety across numerous industries.

In industrial applications, vibration sensors play a significant role in predictive maintenance. Machines like motors, pumps, turbines, and compressors produce unique vibration patterns during normal operation. Any deviation from these patterns may signal wear and tear, imbalance, or potential malfunction. By embedding vibration sensors into equipment, industries can monitor machinery health continuously, reducing unplanned downtime and extending the lifespan of expensive assets. Furthermore, vibration analysis allows for cost savings by enabling maintenance only when necessary, rather than following rigid schedules. This condition-based maintenance approach has become a cornerstone of Industry 4.0 practices.

Beyond industry, vibration sensors are widely used in structural health monitoring. Bridges, buildings, and other infrastructures face stress from natural forces like wind, earthquakes, and traffic. By deploying vibration sensors in these structures, engineers can assess their stability, detect cracks, and prevent catastrophic failures. In seismic studies, vibration sensors detect ground movements, contributing to earthquake monitoring and disaster preparedness. These applications highlight the role of vibration sensing in public safety and risk management.

In the field of consumer electronics, vibration sensors are integrated into mobile devices, gaming controllers, and wearables to enhance user experience. For example, accelerometers in smartphones enable functions like screen rotation, step counting, and motion-based gaming. In automotive systems, vibration sensors contribute to driver assistance features, crash detection, and engine monitoring. Their compact size and low power consumption make them suitable for portable and embedded systems.

The working principle of vibration sensors varies depending on their design. Piezoelectric sensors generate electric charge when subjected to mechanical stress, making them suitable for high-frequency vibration measurement. Accelerometers, on the other hand, measure acceleration forces that can be static or dynamic, thereby detecting vibration as changes in velocity over time. MEMS (Micro-Electro-Mechanical Systems) sensors leverage microfabrication technology to create small, sensitive devices widely used in modern

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electronics. These diverse mechanisms ensure that vibration sensors can address the needs of both high-precision scientific instruments and cost-effective consumer devices.

In vibration sensors are versatile and indispensable tools in today's technology-driven world. Their ability to capture and interpret mechanical oscillations underpins critical systems in industries, infrastructure, consumer electronics, and safety applications. As technology advances, vibration sensors are becoming smarter, more energy-efficient, and capable of integration with IoT platforms. This evolution will expand their relevance in predictive analytics, smart cities, and autonomous systems, reinforcing their position as key enablers of innovation and safety in diverse domains.

3.1.3 Webcam

A webcam is a digital video capture device that connects to computers or networks to stream or record live visual content. Initially designed as peripheral devices for personal computers, webcams have evolved into integral components of laptops, tablets, smartphones, and smart appliances. They serve as the primary interface for real-time communication, enabling video conferencing, online education, remote work, and live streaming. By converting optical input into digital signals, webcams make it possible to transmit visual information across global networks instantly. With the rise of digital communication, webcams have become essential not only for personal use but also for professional, educational, and industrial purposes.

The basic functioning of a webcam involves a lens system that captures light from the environment, which is then processed by an image sensor such as CCD (Charge-Coupled Device) or CMOS (Complementary Metal-Oxide-Semiconductor). The sensor converts light into electrical signals, which are digitized and transmitted through USB, wireless, or integrated computer interfaces. Modern webcams support high-definition (HD) and even 4K video resolutions, providing clear and sharp images for professional applications. Features such as autofocus, low-light correction, and high frame rates enhance the quality of visual communication, making webcams more reliable and user-friendly.

Webcams play a crucial role in communication technologies. Platforms like Zoom, Microsoft Teams, and Google Meet rely heavily on webcam functionality to facilitate remote collaboration. During the global shift toward online learning and remote work, webcams

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emerged as indispensable tools for maintaining connectivity and productivity. In addition to professional communication, webcams are widely used in social media and content creation. Live streaming platforms like YouTube, Twitch, and Instagram leverage webcams for user engagement, enabling individuals to broadcast in real time to global audiences. This shift has transformed webcams from simple accessories into powerful enablers of digital culture and online identity.

Another significant application of webcams is in security and surveillance. Webcams can act as IP cameras, providing continuous monitoring of homes, offices, and public spaces. Integrated with motion detection, night vision, and cloud storage, they enhance security systems by offering real-time video feeds accessible from remote locations. In smart home setups, webcams connect with IoT platforms to provide intelligent surveillance, allowing users to receive alerts and monitor spaces via smartphones. The affordability and accessibility of webcams have democratized surveillance technology, making it available to everyday users.

In education, webcams have transformed the way knowledge is delivered and consumed. Virtual classrooms, online tutoring, and remote laboratories all depend on webcam-enabled platforms to provide interactive and personalized learning experiences. Similarly, in healthcare, telemedicine relies on webcams to connect patients with doctors for remote consultations, improving access to medical services in underserved regions. These applications demonstrate the webcam's versatility in addressing critical societal needs.

In the webcam is more than just a digital camera; it is a gateway to communication, security, education, and creativity. From simple low-resolution devices to advanced AI-powered cameras with background removal and facial recognition, webcams continue to evolve in response to technological and social demands. Their ubiquity across personal and professional contexts underscores their importance in the modern digital era, and as network speeds and AI technologies improve, webcams will only become more powerful tools for connection and collaboration.

3.1.4 PIR Sensor

A Passive Infrared (PIR) sensor is a widely used electronic device designed to detect motion by sensing infrared radiation emitted by objects in its field of view. Every object with

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a temperature above absolute zero emits infrared radiation, and humans and animals, being warm-blooded, emit significant amounts of it. PIR sensors operate by measuring changes in infrared radiation levels, allowing them to detect movement without the need for physical contact or active signal emission. This non-intrusive and energy-efficient detection makes PIR sensors ideal for applications such as security systems, automation, and energy conservation.

The working principle of a PIR sensor involves the use of pyroelectric materials, which generate an electrical signal when exposed to varying levels of infrared radiation. The sensor is divided into zones, and when a warm body moves across these zones, it causes a change in the detected infrared pattern, resulting in a voltage output. This signal is then processed by control circuits to trigger actions such as activating lights, sounding alarms, or sending notifications. PIR sensors are called "passive" because they do not emit radiation themselves but only detect emitted radiation, distinguishing them from active sensors like ultrasonic or radar sensors.

PIR sensors are most commonly associated with security and intrusion detection systems. They are widely installed in homes, offices, and industrial sites to detect unauthorized movement and trigger alarms or surveillance cameras. Their reliability, low cost, and low power consumption make them a preferred choice in security solutions. In addition, they are often combined with other sensors and technologies to enhance accuracy and reduce false alarms caused by environmental factors such as heat sources or sunlight.

Another key application of PIR sensors is in energy management. They are frequently used in automatic lighting systems, where lights turn on only when motion is detected and turn off after a period of inactivity. This reduces unnecessary energy consumption in homes, offices, and public buildings, contributing to cost savings and environmental sustainability. Similarly, PIR sensors are used in HVAC (heating, ventilation, and air conditioning) systems to regulate temperature and airflow based on room occupancy, improving comfort while conserving energy.

In healthcare and eldercare, PIR sensors play a role in monitoring patient movement and ensuring safety. For instance, they can detect when a patient leaves their bed or moves within a monitored area, allowing caregivers to respond promptly. In smart homes, PIR sensors are integrated into automation systems to provide convenience and enhance the user

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experience. They enable intelligent control of devices such as doors, appliances, and entertainment systems based on user presence.

In PIR sensors are versatile, cost-effective, and energy-efficient devices that serve as the backbone of many motion detection systems. Their passive detection mechanism, combined with simplicity and reliability, makes them suitable for a wide range of applications spanning security, automation, healthcare, and energy management. As IoT and smart technologies continue to expand, PIR sensors will remain crucial components, offering seamless interaction between humans and their environment while contributing to smarter, safer, and more sustainable living spaces.

3.1.5 Ultrasonic Sensor

The ultrasonic sensor is an electronic device designed to measure distance by emitting ultrasonic waves and calculating the time it takes for the echo to return after hitting an object. Typically operating at frequencies above human hearing, such as 40 kHz, these sensors provide accurate and real-time distance measurements across short to medium ranges. The sensor usually consists of a transmitter that emits sound pulses and a receiver that detects the returning echo. Using the speed of sound in air, the device calculates the distance to the object, making it highly reliable for non-contact detection. The simplicity of its working principle combined with low cost and durability has made ultrasonic sensors extremely popular in automation and robotics.

Fig 3.3 Ultrasonic sensor

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One of the main strengths of ultrasonic sensors is their independence from visual conditions such as lighting, color, or transparency of the object. Unlike infrared sensors, they can detect objects with varying surface textures and even operate in complete darkness. This makes them useful in a wide variety of applications, including robotics for obstacle avoidance, automotive systems for parking assistance, industrial automation for level measurement, and healthcare for patient monitoring. Their ruggedness allows them to function in dusty, humid, or foggy environments, which might impair optical sensors. Additionally, the power consumption of ultrasonic sensors is minimal, making them suitable for battery-operated systems and continuous monitoring tasks.

In practical usage, ultrasonic sensors are often paired with microcontrollers for processing and decision-making. They can provide digital or analog outputs depending on the design, and their integration with Arduino, Raspberry Pi, or other platforms is straightforward. Calibration and positioning are essential to ensure accuracy, as environmental factors like temperature and airflow may slightly affect sound propagation. With advancements in sensor technology, modern ultrasonic devices come with improved range, resolution, and response times, expanding their usability in fields such as drones, smart vehicles, and automated industrial lines. Overall, their affordability, reliability, and versatility make them an indispensable tool in sensing and measurement technologies.

3.1.6 GSM

The Global System for Mobile Communications (GSM) is a standard developed to facilitate reliable and secure mobile voice and data transmission across cellular networks. It operates on multiple frequency bands such as 900 MHz, 1800 MHz, and 1900 MHz, enabling worldwide adoption and interoperability. GSM technology employs digital transmission and time division multiple access (TDMA) to allocate frequency bands among multiple users efficiently. This ensures that communication remains stable, even when multiple devices are active within the same network area. Over the years, GSM has become the backbone of mobile communication, enabling not only voice calls but also SMS (Short Message Service), data transmission, and roaming capabilities across countries. Its global standardization and extensive infrastructure have made GSM modules a reliable choice in embedded systems and IoT projects.

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Fig 3.4 GSM Module

One of the significant advantages of GSM is its integration capability with microcontrollers and embedded devices. GSM modules such as SIM800 or SIM900 allow developers to add SMS, voice, and GPRS data features to their projects. For example, a microcontroller can send AT commands to a GSM module, enabling the system to communicate alerts, transmit sensor data, or even receive commands remotely. This has made GSM highly valuable in applications like home automation, industrial monitoring, fleet management, and emergency systems. Unlike short-range communication technologies such as Bluetooth or Wi-Fi, GSM leverages cellular infrastructure, allowing global communication as long as a SIM card with network coverage is available. This feature makes GSM particularly effective for systems that require long-distance communication and monitoring.

In practical applications, GSM technology has expanded beyond mobile phones to serve as a critical communication medium in embedded solutions. Its robustness ensures reliable operation in remote areas where other wireless networks may be unavailable. GSM-based alert systems, for instance, are commonly used in security devices, weather monitoring stations, and healthcare solutions to notify users or authorities during emergencies. As technology has advanced, GSM has been integrated with GPS for location tracking, providing real-time monitoring of vehicles or assets. Although newer cellular technologies such as 4G and 5G are gradually replacing GSM, its simplicity, affordability, and reliability ensure it continues to be widely used in IoT applications. Its ability to provide uninterrupted connectivity with minimal infrastructure has cemented GSM’s role as a cornerstone in wireless communication.

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3.1.6 GPS

The Global Positioning System (GPS) is a satellite-based navigation technology that enables accurate determination of geographic location anywhere on Earth. GPS operates by using a network of satellites that transmit signals to receivers, which then calculate the user's position based on the time taken for signals to travel. The system provides latitude, longitude, altitude, and time data, making it indispensable in navigation, mapping, and tracking applications. Initially developed for military use, GPS has become a global utility for both civilian and commercial applications. With an accuracy that can range from a few meters to centimeters (when combined with advanced correction techniques), GPS has revolutionized fields such as transportation, agriculture, disaster management, and logistics.

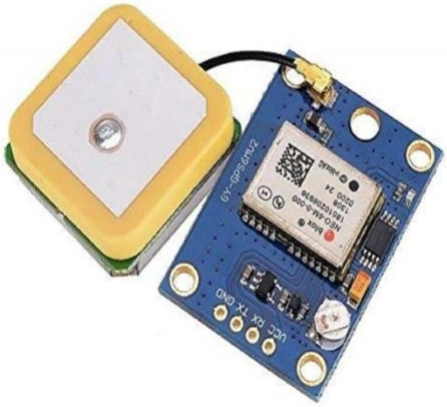

Fig 3.5 GPS Module

A significant advantage of GPS is its versatility and reliability in real-time location tracking. Whether integrated into smartphones, vehicles, or IoT devices, GPS provides continuous updates on position, which can be displayed on maps or transmitted to monitoring systems. This has enabled advancements in applications such as ride-hailing services, personal navigation devices, and geofencing-based security systems. Moreover, industries such as precision agriculture rely heavily on GPS for tasks like automated planting, irrigation, and harvesting. Aviation, shipping, and railways also depend on GPS for safe and efficient navigation. Its integration with mobile communication networks allows for enhanced tracking, where data can be transmitted to remote servers for monitoring and analysis.

Modern GPS receivers are compact, energy-efficient, and capable of working with multiple satellite constellations such as GLONASS (Russia), Galileo (Europe), and BeiDou

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(China), improving accuracy and reliability. GPS modules such as NEO-6M or u-blox are widely used in embedded projects, interfacing seamlessly with microcontrollers via UART or I²C communication. These modules enable developers to design smart systems for fleet tracking, personal safety, environmental monitoring, and asset management. Challenges such as signal obstruction in tunnels or dense urban environments are being mitigated by integrating GPS with inertial navigation systems or cellular triangulation. With continuous advancements, GPS continues to play a critical role in enabling real-time, location-based services, driving innovation in both consumer and industrial domains.

LCD (LIQUID CRYSTAL DISPLAY)

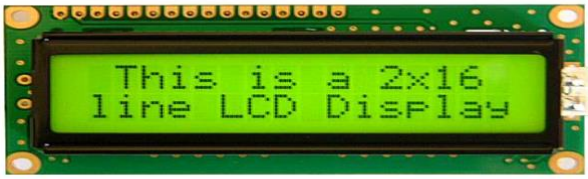

Fig 3.6 LCD Display

3.1.7 LCD Display

LCD (Liquid Crystal Display) screen is an electronic display module and find a wide range of applications. A 16x2 LCD display is very basic module and is very commonly used in various devices and circuits. A 16x2 LCD means it can display 16 characters per line and there are 2 such lines. In this LCD each character is displayed in 5x7 pixel matrix. This LCD has two registers, namely, Command and Data. The command register stores the command instructions given to the LCD, The data register stores the data to be displayed on the LCD.

The data is the ASCII value of the character to be displayed on the LCD.Liquid crystal displays are used for display of numeric and alphanumeric character in dot matrix and segmental displays. The two liquid crystal materials which are commonly used in display technology are nematic and cholesteric whose schematic arrangement of molecules is shown in fig. The most popular liquid crystal structure is the Nematic Liquid Crystal (NLC). In this all the molecules align themselves approximately parallel to a unique axis (director), while

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retaining the complete translational freedom. The liquid is normally transparent, but if subjected to a strong electric field, disruption of the well ordered crystal structure takes place causing the liquid to polarize and turn opaque. The removal of the applied electric field allows the crystal structure to regain its original form and the materials become transparent.

Based on the construction, LCD's are classified into two types. They are,

(i) Dynamic scattering type

(ii) Field effect type.

Dynamic scattering type

The construction of the dynamic scattering liquid crystal cell is shown in the fig. The display consists of two glass plates, each coated with tin oxide (SnO2) on the inside with transparent electrodes separated by a liquid crystal layer,5μA to 50 μA thick. The oxide coating on the front sheet is etched to produce a single or multi- segment pattern of characters, with each segment properly insulated from each other. A weak electric field applied to liquid crystal tends to align molecule in the direction of the field. As soon as the voltage exceeds certain threshold value, the domain structure collapses and the appearance is changed. As the voltage grows further, the flow becomes turbulent and the substance turns optically homogenous. In this disordered state, the liquid crystal scatters light.

Thus, when the liquid is not activated, it is transparent. When the liquid is activated, the molecular turbulence causes light to be scattered in all directions and the cell appears bright. This phenomenon is called dynamic scattering

Field effect type

The construction of the field effect LCD display is similar to that of the dynamic scattering type, with the expectation that two thin polarizing optical filters are placed at the inside of each glass sheet.

The LCD material is of twisted nematic type which twists the light (change in direction of polarization) passing through the cell when the latter is not energized. This allows light to pass through the optical filters and the cell appears bright. When the cell is energized, no twisting of light takes place and the cell appears dull.

Liquid crystal cells are of two types (i) transitive type (ii) reflective type. In the transitive type cell both glass sheets are transparent so that the light from the rear source is

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scattered in the forward direction when the cell is activated. The reflecting type cell has a reflecting surface on one side of the glass sheet. The incident light on the front surface of the cell is dynamically scattered by an activated cell. Both types of cells appear quite bright when activated even under ambient light conditions.

Liquid crystals consume small amount of energy, in a seven segment display the current drawn is about 25 µA for dynamic scattering cells and 300 µA .for field effect cells LCD's require ac voltage supply. A typical voltage supply to dynamic scattering LCD's are normally used for seven-segmental displays.

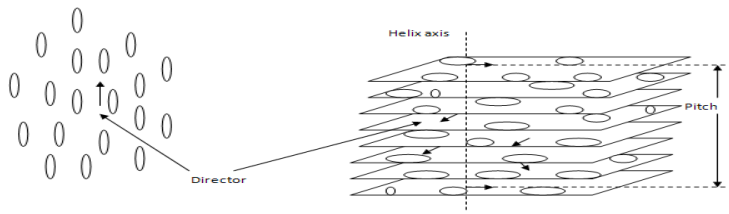

Fig 3.7 Schematic arrangement in liquid crystal

3.1.8 Power Supplies

A power supply (sometimes known as a power supply unit or PSU) is a device or system that supplies electrical or other types of energy to an output load or group of loads. The term is most commonly applied to electrical energy supplies, less often to mechanical ones, and rarely to others.

This circuit is a small +5V power supply, which is useful when experimenting with digital electronics. Small inexpensive wall transformers with variable output voltage are available from any electronics shop and supermarket. Those transformers are easily available, but usually their voltage regulation is very poor, which makes then not very usable for digital circuit experimenter unless a better regulation can be achieved in some way. The following circuit is the answer to the problem.

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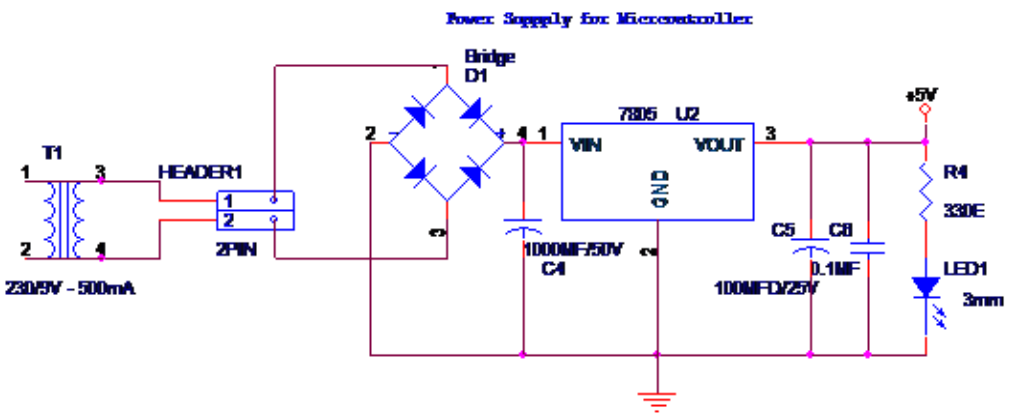

Fig 3.8 Block diagram of power supply

3.1.9 Transformer

A transformer is a device that transfers electrical energy from one circuit to another through inductively coupled wires. A changing current in the first circuit (the primary) creates a changing magnetic field; in turn, this magnetic field induces a changing voltage in the second circuit (the secondary). By adding a load to the secondary circuit, one can make current flow in the transformer, thus transferring energy from one circuit to the other. The secondary induced voltage V_S is scaled from the primary V_P by a factor ideally equal to the ratio of the number of turns of wire in their respective windings:

$$\frac{V_S}{V_P} = \frac{N_S}{N_P}$$

By appropriate selection of the numbers of turns, a transformer thus allows an alternating voltage to be stepped up — by making N_S more than N_P or stepped down, by making it less.

A key application of transformers is to reduce the current before transmitting electrical energy over long distances through wires. Most wires have resistance and so dissipate electrical energy at a rate proportional to the square of the current through the wire. By transforming electrical power to a high-voltage, and therefore low-current form for transmission and back again afterwards, transformers enable the economic transmission of power over long distances. Consequently, transformers have shaped the electricity supply industry, permitting generation to be located remotely from points of demand. All but a fraction of the world's electrical power has passed through a series of transformers by the time it reaches the consumer.

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Transformers are some of the most efficient electrical 'machines', with some large units able to transfer 99.75% of their input power to their output. Transformers come in a range of sizes from a thumbnail-sized coupling transformer hidden inside a stage microphone to huge gigavolt-ampere-rated units used to interconnect portions of national power grids. All operate with the same basic principles, though a variety of designs exist to perform specialized roles throughout home and industry.

The transformer is based on two principles: first, that an electric current can produce a magnetic field (electromagnetism) and, second, that a changing magnetic field within a coil of wire induces a voltage across the ends of the coil (electromagnetic induction). By changing the current in the primary coil, one changes the strength of its magnetic field; since the secondary coil is wrapped around the same magnetic field, a voltage is induced across the secondary.

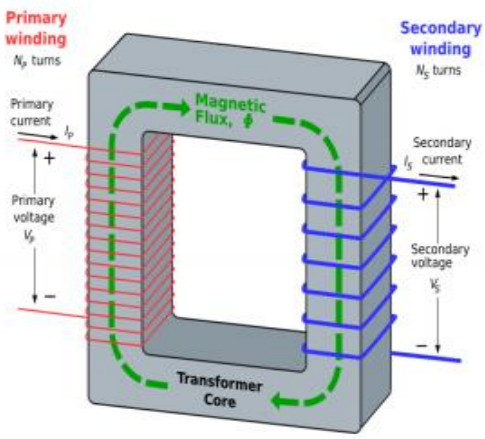

Fig 3.9 An ideal step-down transformer

A simplified an ideal step-down transformer design is shown in the above figure. A current passing through the primary coil creates a magnetic field. The primary and secondary coils are wrapped around a core of very high magnetic permeability, such as iron; this ensures that most of the magnetic field lines produced by the primary current are within the iron and pass through the secondary coil as well as the primary coil.

3.1.10 Rectifier

A rectifier is an electrical device that converts alternating current (AC), which periodically reverses direction, to direct current (DC), which flows in only one direction. The process is known as rectification. Rectifiers are used as components of power supplies and as

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detectors of radio signals. Mainly there are three types of rectifier i.e. half wave rectifier, full wave rectifier and Bridge Rectifier.

Half-wave rectifier

In half-wave rectification of a single-phase supply, either the positive or negative half of the AC wave is passed, while the other half is blocked. Because only one half of the input waveform reaches the output, mean voltage is lower. Half-wave rectification requires a single diode in a single-phase supply, or three in a three-phase supply. Rectifiers yield a unidirectional but pulsating direct current; half-wave rectifiers produce far more ripple than full-wave rectifiers, and much more filtering is needed to eliminate harmonics of the AC frequency from the output.

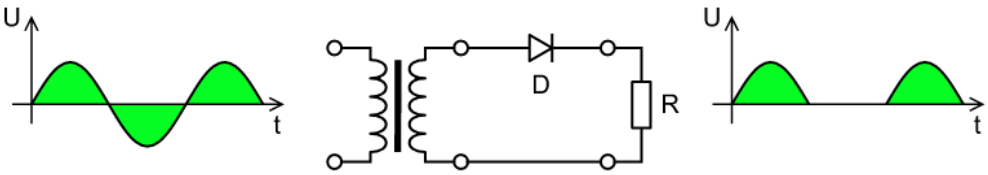

Fig 3.10 Half Wave Rectifier

Full-wave rectifier

A full-wave rectifier converts the whole of the input waveform to one of constant polarity (positive or negative) at its output. Full-wave rectification converts both polarities of the input waveform to pulsating DC (direct current), and yields a higher average output voltage. Two diodes and a center tapped transformer are needed.

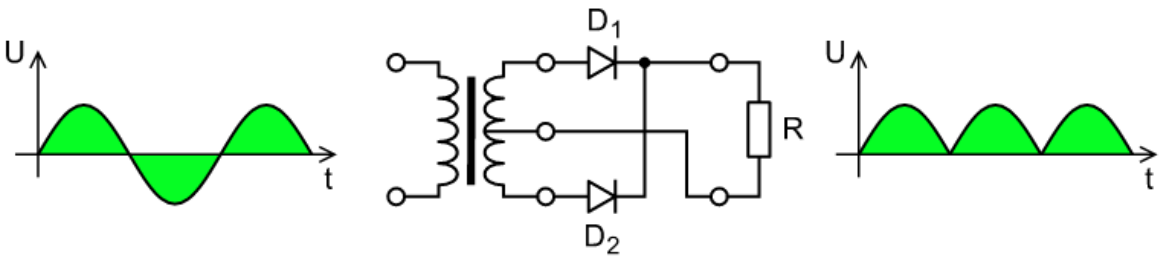

Fig 3.11 Full-Wave Rectifier

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Bridge Rectifier

A diode bridge is an arrangement of four (or more) diodes in a bridge circuit configuration that provides the same polarity of output for either polarity of input. When used in its most common application, for conversion of an alternating current (AC) input into a direct current (DC) output, it is known as a bridge rectifier. A bridge rectifier provides full-wave rectification from a two-wire AC input, resulting in lower cost and weight as compared to a rectifier with a 3-wire input from a transformer with a center-tapped secondary winding. The essential feature of a diode bridge is that the polarity of the output is the same regardless of the polarity at the input.

Basic operation

According to the conventional model of current flow, current is defined to be positive when it flows through electrical conductors from the positive to the negative pole. In actuality, free electrons in a conductor nearly always flow from the negative to the positive pole. In the vast majority of applications, however, the actual direction of current flow is irrelevant. Therefore, in the discussion below the conventional model is retained.

In the diagrams below, when the input connected to the left corner of the diamond is positive, and the input connected to the right corner is negative, current flows from the upper supply terminal to the right along the red (positive) path to the output, and returns to the lower supply terminal via the blue (negative) path.

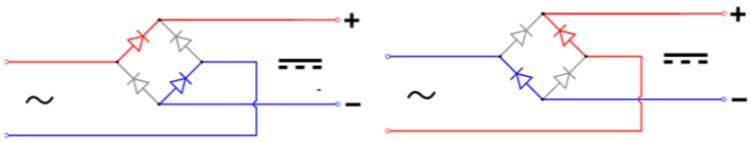

Fig 3.12 Operation of bridge rectifier

When the input connected to the left corner is negative, and the input connected to the right corner is positive, current flows from the lower supply terminal to the right along the red (positive) path to the output, and returns to the upper supply terminal via the blue (negative) path.

In each case, the upper right output remains positive and lower right output negative. Since this is true whether the input is AC or DC, this circuit not only produces a DC output from an AC input, it can also provide what is sometimes called "reverse polarity protection".

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That is, it permits normal functioning of DC-powered equipment when batteries have been installed backwards, or when the leads (wires) from a DC power source have been reversed, and protects the equipment from potential damage caused by reverse polarity.

IC Voltage Regulators

Voltage regulators comprise a class of widely used ICs. Regulator IC units contain the circuitry for reference source, comparator amplifier, control device, and overload protection all in a single IC. Although the internal construction of the IC is somewhat different from that described for discrete voltage regulator circuits, the external operation is much the same. IC units provide regulation of either a fixed positive voltage, a fixed negative voltage, or an adjustable set voltage. A power supply can be built using a transformer connected to the ac supply line to step the ac voltage to desired amplitude, then rectifying that ac voltage, filtering with a capacitor and RC filter, if desired, and finally regulating the dc voltage using an IC regulator. The regulators can be selected for operation with load currents from hundreds of milliamperes to tens of amperes, corresponding to power ratings from milliwatts to tens of watts.

Three-Terminal Voltage Regulators

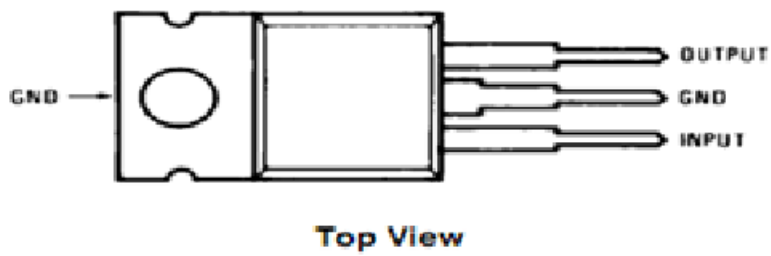

Fig 3.13 Three-Terminal Voltage Regulators

Figure shows the basic connection of a three-terminal voltage regulator IC to a load. The fixed voltage regulator has an unregulated dc input voltage, V_i, applied to one input terminal, a regulated output dc voltage, V_o, from a second terminal, with the third terminal connected to ground. For a selected regulator, IC device specifications list a voltage range over which the input voltage can vary to maintain a regulated output voltage over a range of load current. The specifications also list the amount of output voltage change resulting from a change in load current (load regulation) or in input voltage (line regulation). The series 78 regulators provide fixed regulated voltages from 5 to 24 V. Figure shows how one such IC, a

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7805, is connected to provide voltage regulation with output from this unit of +5V dc. An unregulated input voltage V is filtered by capacitor C1 and connected to the IC's IN terminal. The IC's OUT terminal provides a regulated + 12V which is filtered by capacitor C2 (mostly for any high-frequency noise). The third IC terminal is connected to ground (GND). While the input voltage may vary over some permissible voltage range, and the output load may vary over some acceptable range, the output voltage remains constant within specified voltage variation limits. These limitations are spelled out in the manufacturer's specification sheets. There are two types of voltage regulator they are 78xx series and 79xx series.

78xx series

There are common configurations for 78xx ICs, including 7805 (5 V), 7806 (6 V), 7808 (8 V), 7809 (9 V), 7810 (10 V), 7812 (12 V), 7815 (15 V), 7818 (18 V), and 7824 (24 V) versions. The 7805 is the most common, as its regulated 5-volt supply provides a convenient power source for most TTL components.

Less common are lower-power versions such as the LM78Mxx series (500 mA) and LM78Lxx series (100 mA) from National Semiconductor. Some devices provide slightly different voltages than usual, such as the LM78L62 (6.2 volts) and LM78L82 (8.2 volts) as well as the STMicroelectronics L78L33ACZ (3.3 volts).

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Python 3.7.4

Python is a general-purpose interpreted, interactive, object-oriented, and high-level programming language. It was created by Guido van Rossum during 1985- 1990.

Fig 3.14 Python

Like Perl, Python source code is also available under the GNU General Public License (GPL). This tutorial gives enough understanding on Python programming language.

Python is a high-level, interpreted, interactive and object-oriented scripting language. Python is designed to be highly readable. It uses English keywords frequently where as other languages use punctuation, and it has fewer syntactical constructions than other languages. Python is a MUST for students and working professionals to become a great Software Engineer specially when they are working in Web Development Domain.

Python is currently the most widely used multi-purpose, high-level programming language. Python allows programming in Object-Oriented and Procedural paradigms. Python programs generally are smaller than other programming languages like Java. Programmers have to type relatively less and indentation requirement of the language, makes them readable all the time. Python language is being used by almost all tech-giant companies like – Google, Amazon, Facebook, Instagram, Dropbox, Uber… etc. The biggest strength of Python is huge collection of standard libraries which can be used for the following:

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- Machine Learning
- GUI Applications (like Kivy, Tkinter, PyQt etc.)
- Web frameworks like Django (used by YouTube, Instagram, Dropbox)
- Image processing (like OpenCV, Pillow)
- Web scraping (like Scrapy, BeautifulSoup, Selenium)
- Test frameworks
- Multimedia
- Scientific computing
- Text processing and many more.

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REVIEW OF LITERATURE

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4. REVIEW OF LITERATURE

Dmitry Yudin [1] proposed a system for the detection of big animals on images with road scenes using deep learning, aiming to reduce vehicle-animal collisions by improving visibility and early detection. The framework leverages convolutional neural networks (CNNs) to analyze roadside images captured from cameras installed near highways. These CNN models were trained on diverse datasets containing large animals such as cattle, deer, and wild species in varying road and weather conditions. By learning distinct animal features, the system achieves high classification accuracy while minimizing false alarms. The use of deep learning ensures robustness compared to traditional image processing techniques, which often fail under poor lighting or cluttered backgrounds. The research highlights the scalability of CNN models for real-world deployment across different regions and traffic scenarios. Furthermore, the experimental results demonstrated strong adaptability in both day and night environments. This study emphasizes the importance of automated image-based systems in ensuring road safety. By providing real-time analysis, the framework offers timely alerts to drivers, reducing accidents caused by sudden animal crossings. Overall, the research positions deep learning as a reliable solution for integrating vision-based detection into intelligent transportation systems.

Saravanan Alagarsamy [2] introduced a framework for the detection of animals for road safety using deep learning, focusing on providing drivers with early warnings to prevent accidents. The system employs advanced object detection architectures such as YOLO to perform both localization and recognition of animals in dynamic traffic environments. Unlike traditional techniques, YOLO enables the simultaneous identification of multiple animals within a single frame while maintaining real-time speed. The model was trained with road scene datasets covering different animal species and environmental variations, ensuring robustness. By extracting features effectively and minimizing false detections, the system achieves higher precision and recall compared to conventional detection methods. The research also emphasizes the importance of deploying lightweight models that can run efficiently on embedded hardware. Experimental outcomes demonstrated strong adaptability in rural and forest-adjacent highways where animal crossings are frequent. The study underscores the role of deep learning in balancing accuracy with real-time performance for

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practical road safety solutions. Overall, the proposed method delivers a scalable, low-latency detection framework that can be integrated into intelligent driver assistance systems.

S. Meera [3] developed an animal detection alert system designed to improve driver awareness and reduce collisions caused by sudden animal crossings. The approach integrates sensor-based monitoring with image analysis to continuously observe roadside environments. When the system detects an animal entering a danger zone, immediate alerts are generated through visual or auditory signals. The detection pipeline employs machine learning algorithms trained to distinguish animals from other moving objects, thereby minimizing false alarms. This integration of detection with alert mechanisms enhances the practical usability of the system, as drivers can respond quickly to potential hazards. The research emphasizes cost-effectiveness by utilizing affordable sensors and cameras that are easy to deploy along highways. By combining real-time monitoring with an automated alert framework, the system ensures timely preventive action. Experimental trials showed improved response times in accident-prone zones. This work highlights the potential of multi-modal approaches that combine detection with immediate feedback, bridging the gap between technology and driver safety. Overall, the system represents a practical, real-world application of animal detection in rural and highway road safety initiatives.

K. Parkavi [4] proposed a system aimed at *enhancing road safety by detecting animals on highways during night,* addressing the unique challenges of low-visibility environments. The approach combines infrared imaging with deep learning-based detection algorithms to identify animals under dark and poorly lit conditions. Traditional visible-light cameras fail in such settings, but infrared sensors capture heat signatures emitted by warm-blooded animals, providing reliable inputs for classification models. A convolutional neural network (CNN) was used to process these infrared images, learning distinct patterns associated with animals compared to background noise. The study highlights how the fusion of infrared sensing and AI can significantly reduce nighttime accidents, a common problem in rural areas. The system demonstrated high detection accuracy in experimental setups simulating different night environments. By offering timely warnings, it improves driver reaction times and overall road safety. The research also shows scalability for integration with vehicle-mounted cameras and roadside units. Ultimately, this work emphasizes how combining advanced sensing technologies with AI provides effective solutions for persistent road safety challenges in nighttime driving.

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Sachin Sharma [5] focused on automated detection of animals in the Indian context, addressing region-specific challenges such as diverse species, frequent crossings, and highly varied road conditions. The study highlights that most global detection systems fail in Indian roads due to lack of localized data. To address this, a custom dataset of Indian road scenes containing animals like cows, goats, and stray animals was developed. Using machine learning classifiers, the system was trained to recognize these animals in real time while filtering out irrelevant roadside objects. The framework achieved strong detection accuracy despite complex backgrounds such as crowded roads and rural pathways. By tailoring the dataset and algorithm to Indian conditions, the research demonstrates the necessity of localized solutions for effective deployment. The system's lightweight design allows for deployment in embedded devices or roadside units. The findings underscore the adaptability of machine learning techniques to specific geographical contexts. This research contributes to reducing accidents in India, where animal-related collisions are a significant concern. Overall, the study proves that contextualized datasets are vital for building effective and reliable road safety technologies.

S. Simonthomas [6] proposed a machine learning-based wild-animal detection system near roads using IoT sensors and optimization techniques. Unlike vision-only methods, this framework integrates sensor data with learning algorithms to improve detection accuracy and reliability. IoT-enabled devices deployed near highways capture environmental parameters such as movement and heat signatures, which are then analyzed using optimized machine learning models. The optimization process reduces computational overhead and energy usage, making the system suitable for long-term deployment in resource-constrained environments. The study emphasizes the synergy of IoT and AI for real-world safety applications. By processing data locally and transmitting only essential alerts, the system ensures efficiency while maintaining responsiveness. Experimental results showed that combining sensor inputs with AI reduces false alarms compared to image-only approaches. This hybrid solution provides flexibility for deployment in areas where camera visibility is limited by terrain or weather. The research highlights the importance of cross-technology integration to overcome single-approach limitations. Overall, the work provides a scalable and intelligent framework for reducing wildlife-vehicle collisions.

Pravin A [7] introduced a vision-based technique for animal detection to minimize accidents on highways and rural roads. The method employs image processing techniques

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such as edge detection, segmentation, and contour analysis to isolate animals from backgrounds in video streams. These extracted features are then classified using machine learning models to determine whether an object is an animal. Unlike sensor-based systems, vision-based approaches are cost-effective and easy to implement with existing surveillance or vehicle cameras. The study shows how basic computer vision can be combined with AI classifiers for improved accuracy. The system was tested under varying conditions, including changing light and background clutter, and achieved satisfactory results for real-time applications. By focusing on vision, the framework reduces dependency on additional hardware, lowering overall system costs. However, the study also notes challenges such as sensitivity to poor lighting and weather conditions. Despite this, the approach demonstrates the potential of vision-driven systems in addressing road safety issues effectively. Overall, the research contributes to scalable and affordable solutions for animal detection.

Prema Arokia Mary G [8] presented a wild animal detection system designed to provide early warnings to drivers on highways near forest regions. The system integrates camera-based inputs with deep learning classifiers to identify animals in real time. Upon detection, both visual and auditory alerts are generated, ensuring the driver is immediately informed of potential danger. The model is trained on datasets containing common wild animals, enabling it to distinguish them from other objects. By combining detection with alerts, the system enhances driver response times, significantly reducing the risk of accidents. The research highlights the practical usability of such systems in regions prone to wildlife crossings. The framework also emphasizes the scalability of integration with vehicle dashboards or roadside infrastructure. Experimental trials demonstrated consistent accuracy across multiple test scenarios. This system underscores the role of combining AI-driven detection with human-centered alert mechanisms to improve road safety. Overall, the study contributes to developing intelligent safety tools for highways in wildlife-rich zones.

Ravishankar S. Bevinal [9] proposed a road friction object detection system aimed at enhancing overall road safety by identifying unexpected objects, including animals, on highways. The approach employs deep learning models trained on diverse datasets containing various roadside objects. Unlike single-purpose systems, this framework provides a broader detection mechanism by classifying multiple types of hazards such as debris, rocks, or animals. The study uses convolutional neural networks (CNNs) to process images and recognize patterns that distinguish obstacles from the road environment. This integrated

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approach reduces false positives by analyzing context and object type. The system was tested under different lighting and weather conditions, demonstrating strong generalization. By detecting animals as part of a broader safety framework, it ensures that road safety measures are holistic and adaptable. The research highlights the value of integrating animal detection into comprehensive obstacle detection systems, improving scalability across different regions. Overall, the study demonstrates how deep learning can deliver robust, real-time, and multi-purpose road safety solutions.

Waleed Saad [10] presented a comprehensive study on loose animal-vehicle accidents mitigation: vision and challenges, focusing on both the technological advancements and the barriers in implementing reliable solutions. The research highlights how loose animal crossings on highways remain a significant contributor to road accidents, especially in rural and forest-adjacent areas. Instead of proposing a single detection model, the work provides a broad vision that integrates computer vision, deep learning, IoT sensors, and vehicular communication technologies to reduce collision risks. The author examines existing approaches such as image-based detection, infrared sensing, and GPS-enabled animal tracking, analyzing their strengths and limitations in real-world deployment. Special emphasis is placed on challenges such as false alarms, poor visibility at night, and the difficulty of scaling solutions across diverse terrains and species. The study also outlines the importance of predictive analytics that can forecast animal movement patterns rather than relying only on real-time detection. By combining prevention strategies with driver alert systems and intelligent transportation networks, the research provides a holistic framework for mitigating accidents. Overall, the study serves as both a roadmap and a critical evaluation of ongoing work in this domain, offering valuable insights for future research on safe and intelligent road ecosystems.

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DEFINITION OF THE PROBLEM AND SCOPE OF THE RESEARCH WORK

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5. DEFINITION OF THE PROBLEM AND SCOPE OF THE RESEARCH WORK

5.1 DEFINITION OF THE PROBLEM

Road accidents caused by the sudden appearance of animals are a growing concern, leading to human injuries, fatalities, property damage, and loss of wildlife. Existing systems for monitoring and detection are either limited to specific vehicle technologies or lack the ability to function effectively in diverse environments, particularly under conditions such as low visibility, high traffic density, or unpredictable animal behavior. The absence of a robust, real-time, and scalable detection mechanism results in delayed driver response and inadequate accident prevention. Therefore, there is a critical need for an intelligent system that can reliably detect and classify animals on roads, operate efficiently across varied scenarios, and provide timely warnings to enhance safety for both humans and animals.

5.2 SCOPE OF THE RESEARCH WORK

The research work of this project focuses on integrating advanced Artificial Intelligence (AI) techniques with Internet of Things (IoT) technologies to address road safety challenges caused by the sudden appearance of animals and humans. The system development involves training AI models for accurate real-time object detection, ensuring that both humans and animals are identified under varying conditions such as low light, high speed, and diverse road environments. Simultaneously, IoT modules including GPS and GSM are implemented to enable seamless communication during emergencies. The research also emphasizes optimizing detection accuracy, minimizing response delay, and ensuring reliable voice alerts for the driver. Furthermore, accident handling is strengthened by automatic SMS notifications containing live location details, ensuring immediate support for affected individuals. By combining AI-based detection with IoT-driven communication, the research work aims to deliver a robust and practical safety solution that reduces collision risks and enhances emergency response effectiveness on modern roadways.

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ARCHITECTURE DESIGN

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6. ARCHITECTURE DESIGN

The architecture of the proposed AI and IoT-based road safety system is designed to provide an end-to-end solution that ensures early detection of humans and animals on the road, driver alerting, and emergency communication in the event of an accident. The system architecture combines hardware components, AI algorithms, and IoT communication modules into a seamless framework that operates in real time to prevent collisions and minimize risks. At the core of the architecture is the sensing and detection layer, where cameras or image sensors are strategically positioned on the vehicle to continuously capture the surrounding road environment. These captured frames are processed by an AI-based object detection algorithm, trained to recognize humans and animals with high accuracy under various conditions. The detection algorithm operates in real time and sends immediate recognition results to the decision-making module. Once a human or animal is detected, the system generates an instant voice alert that is delivered to the driver through an onboard speaker. This real-time interaction between the sensing, detection, and alerting modules forms the first layer of defense against potential accidents.

The second major component of the architecture is the IoT-enabled communication layer, which is activated in case of accidents. Despite the preventive mechanism, collisions may still occur due to driver inattention, environmental conditions, or high-speed impact. To handle such scenarios, the system integrates GPS and GSM modules within the IoT framework. The GPS continuously tracks the real-time location of the vehicle, while the GSM module ensures that data is transmitted through mobile networks to registered contacts. If the system detects a collision or sudden halt suggestive of an accident, the IoT module is triggered to send an automated SMS alert to the family members or emergency responders. This SMS contains critical information, including the vehicle's live location, to enable immediate assistance and rescue operations. This two-layer structure—detection and communication—ensures that the architecture not only prevents accidents but also mitigates the consequences if they occur.

Another crucial aspect of the architecture is the integration of processing units that coordinate between sensing, AI, and IoT components. A microcontroller or embedded processor, such as Raspberry Pi or ESP32, serves as the central processing unit for handling data flow across the system. Once the inference is completed, the output is passed to the

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decision-making module, which determines whether a human or animal is present. Based on this decision, the architecture either triggers a preventive voice alert or initiates accident response communication. This modular and layered processing approach allows the system to achieve both speed and accuracy in real-world deployment scenarios.

The architecture also incorporates a user interface and monitoring framework to enhance usability. A simple dashboard can be integrated on a mobile application or web page, enabling the driver or family members to track system functionality and receive notifications. The IoT layer ensures that accident alerts and location information are stored in a cloud server, which can later be accessed for analysis, reporting, or system improvement. This data storage capability not only helps in monitoring but also supports research and future enhancements by providing valuable insights into accident-prone conditions and system response times.

The design of the architecture takes into account the challenges of real-world deployment, including lighting variations, weather changes, and vehicle speed. To address these, the system architecture emphasizes robustness and adaptability. For instance, infrared or low-light cameras may be incorporated to ensure reliable detection during nighttime driving. Similarly, noise reduction algorithms are embedded in the preprocessing stage to minimize false detections. The IoT communication system is also designed with redundancy, ensuring that if one network fails, alternative pathways can still deliver the critical message. This enhances reliability and ensures that the system can operate in diverse environments without compromising its functionality.

In the architecture of the proposed system integrates three major layers: the sensing and detection layer powered by AI-based vision, the processing and decision-making layer for triggering alerts, and the IoT communication layer for accident response. Each layer is interconnected through a central processing unit that manages data flow and ensures real-time performance. By combining these components into a unified framework, the architecture offers a comprehensive solution that prevents accidents through timely detection and alerts, while also ensuring rapid emergency response through IoT communication. This layered and modular design makes the system highly scalable, adaptable, and effective in addressing the critical problem of animal and human detection on roads. The architecture not only provides immediate safety benefits but also lays the foundation for future enhancements, such as integration with autonomous vehicles, cloud-based analytics, and AI model updates for

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improved accuracy. Ultimately, this architecture design reflects a holistic approach to road safety, ensuring both prevention and protection in a single intelligent framework.

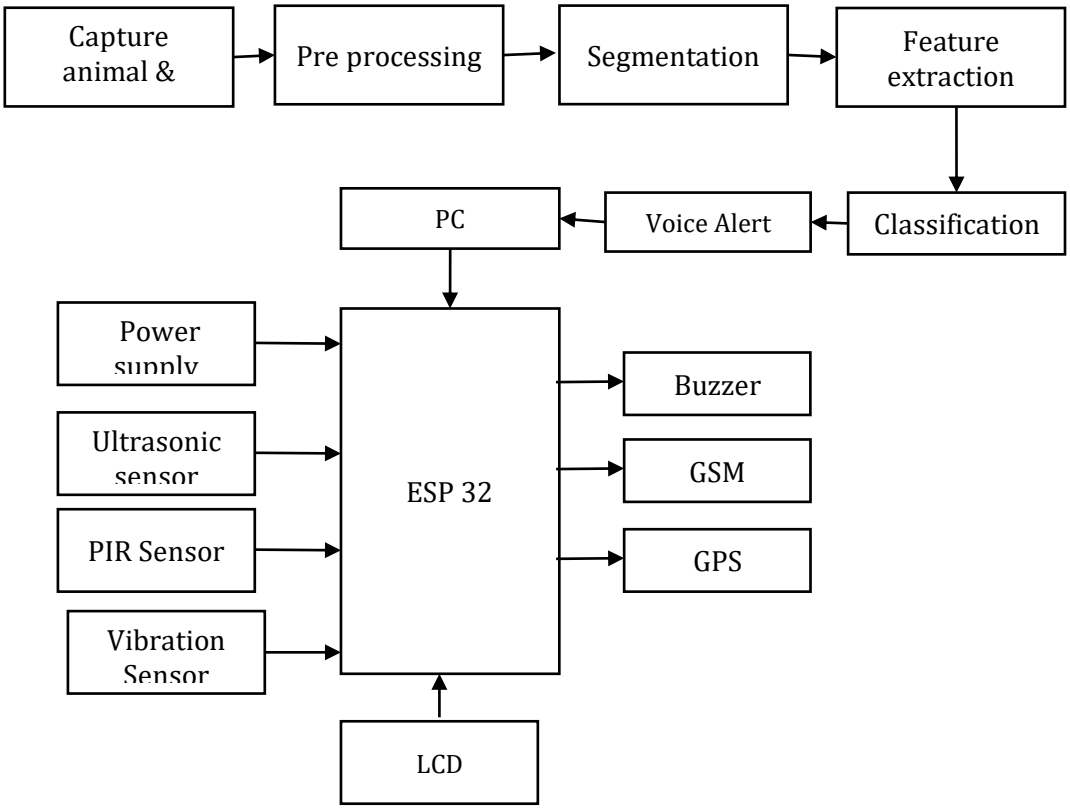

Fig 6.1 System Architecture

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RESULTS AND DISCUSSIONS

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7. RESULTS AND DISCUSSIONS

The development and testing of the AI and IoT-based road safety system demonstrated promising results in real-world and simulated conditions, showing its ability to effectively detect the sudden presence of humans and animals on roads and provide timely alerts to drivers. The system's AI detection module, powered by trained object recognition models, achieved a consistently high detection rate, allowing it to identify potential obstacles with accuracy even in challenging environments. During experimental evaluations, the detection process was completed in real time with very low latency, ensuring that alerts were provided to the driver without noticeable delay. The voice alert mechanism played a crucial role in enhancing driver awareness, as drivers responded more quickly when an audio warning was issued compared to situations without the system. This direct intervention demonstrated how integrating real-time detection with audible alerts can significantly reduce the risk of accidents caused by driver distraction or delayed reaction.

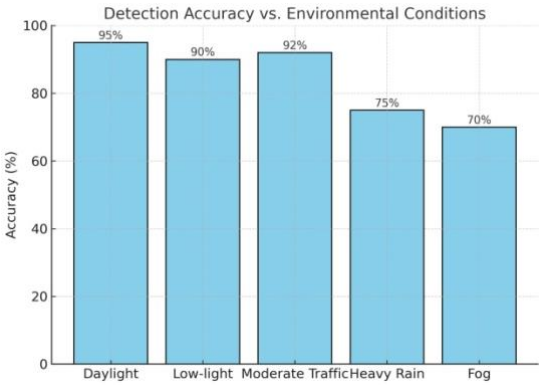

Fig 7.1 Detection Accuracy. Environmental Conditions

In addition to preventive safety, the IoT communication module was evaluated for its reliability in emergency scenarios. Testing showed that, in the event of an accident, the system was able to generate and transmit an SMS alert within a few seconds. This message included the exact GPS location of the vehicle, enabling family members or emergency contacts to quickly identify the accident site and initiate assistance. The speed and accuracy of this communication highlight the practical benefits of combining IoT with road safety systems, particularly in remote areas where accidents often go unnoticed for long periods.

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The architecture also ensured that both detection and communication processes could run simultaneously without interference, maintaining system stability during continuous operation.

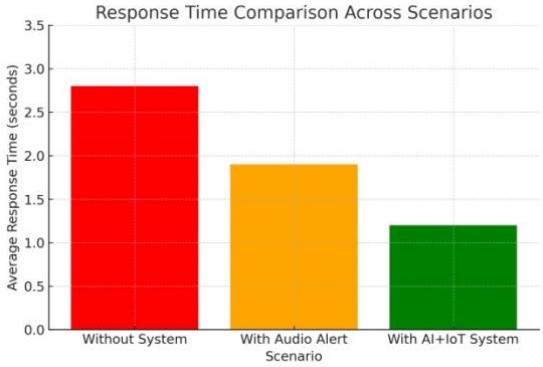

Fig 7.2 Response Time Comparison

Further analysis revealed that the system's performance remained stable under varying conditions such as daylight, low-light, and moderate traffic flow. However, external challenges such as heavy rain, fog, or poor network connectivity were identified as factors that could reduce detection accuracy or delay SMS transmission. These limitations emphasize the need for future enhancements, such as integrating thermal sensors or developing multi-network communication support to overcome environmental and technical constraints. Despite these challenges, the system consistently achieved its intended objectives, proving that AI-based vision combined with IoT connectivity can offer a comprehensive solution to road safety.

The results confirm that the system not only enhances preventive safety through early detection and real-time alerts but also ensures effective post-accident management through automated communication. The discussion of results underlines the reliability, efficiency, and practical applicability of the system in reducing accidents and improving emergency responses. These findings strongly support the system's potential for large-scale deployment and highlight its contribution toward building safer and smarter road transportation.

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CONCLUSION AND FUTURE ENHANCEMENT

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8. CONCLUSION AND FUTURE ENHANCEMENT

8.1 CONCLUSION

In conclusion, the AI and IoT-based road safety system offers a practical and effective solution for minimizing accidents caused by sudden appearances of humans and animals on roads. Its ability to detect obstacles in real time and provide instant voice alerts significantly enhances driver awareness and reduces collision risks. Alongside preventive measures, the IoT module ensures that family members are immediately informed during emergencies through SMS alerts containing accurate GPS location details. The integration of detection, alerting, and communication creates a reliable and efficient mechanism for road safety enhancement. Testing results confirm that the system fulfills its objectives of reducing risks and improving emergency communication, making it suitable for real-world deployment. The lightweight design and use of embedded processors also make the solution affordable and adaptable to different vehicle models. While some challenges remain due to environmental and network constraints, the modular nature of the design allows for future improvements. Overall, the system demonstrates how AI combined with IoT can provide a powerful tool for safer and smarter transportation.

8.2 FUTURE ENHANCEMENT

Future improvements can focus on enhancing detection in adverse conditions such as fog, heavy rain, or nighttime through the integration of infrared or thermal sensors. Expanding the dataset and refining the AI model will increase detection accuracy and adaptability. Cloud-based data storage can be introduced to analyze accident patterns and improve long-term road safety planning. Direct communication with hospitals or emergency services can be added for faster rescue operations. Integration of vehicle-to-vehicle communication can further improve safety by warning nearby vehicles in advance. These enhancements will make the system smarter, more resilient, and aligned with future intelligent transport systems

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